

Patterns of Truth

Location and Modality

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1. Patterns of Truth

Tense logic takes propositions to change truth value across times. Consider the proposition expressed by (1):

(1) There is a major US election.

This proposition is false now, but it will be true on November 3, 2026. Temporary propositions may be true:

- all over a time interval
- entirely within a time interval
- at some of a time interval

Or you may take propositions to change truth value across possible worlds. Consider the proposition expressed by (2):

(2) Someone walked on the Moon in 1969.

This proposition is true in the actual world, but it might have been false had the political will been absent. Contingent propositions may be true:

- all over a region of modal space
- entirely within a region of modal space
- at some of a region of modal space

We aim for a generalization. We leave open whether locations are time intervals, or regions of modal space, or regions of spacetime, what matters is that they enter into the relation of part to whole.

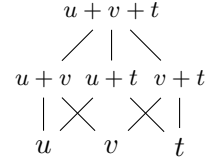
2. Locational Frames

A **locational frame** consists of

- a non-empty set L of locations, and
- a relation of **superlocation** $\sqsupseteq$,

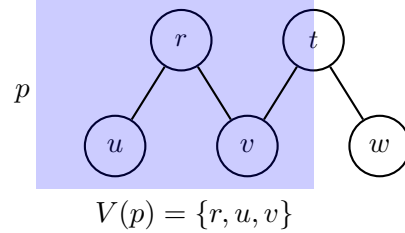
where $u \sqsupseteq v$ means that v is part of u .

$\sqsupseteq$ is reflexive, anti-symmetric, and transitive.



Example. Let u , v , and t be Spring, Summer, and Fall. Spring and Summer lie below Spring + Summer because they are subintervals of it.

Truth at a location. We begin with the relation an atomic proposition bears to a location if it is true *all over* that location.



Heredity. If an atom is true at a location, then it remains true at every sublocation.

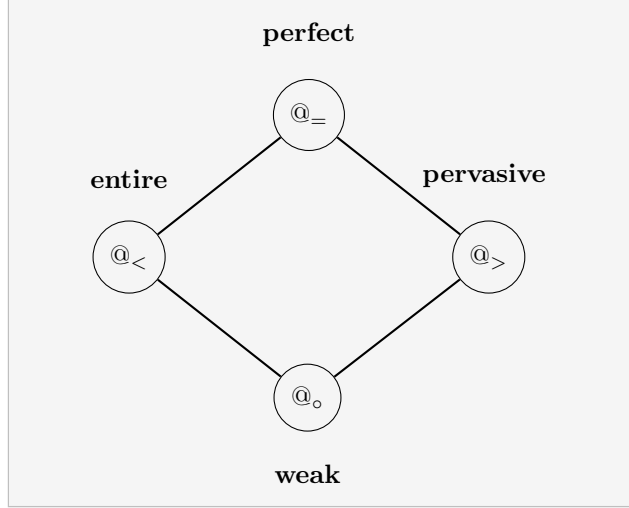
3. Locational Modalities

Two basic modal operators:

- $u \Vdash \Box \varphi$ for all $v \in L$, if $u \sqsupseteq v$, then $v \Vdash \varphi$.
all sublocations verify φ
 φ is true **all over** u
- $u \Vdash \Box \varphi$ for all $v \in L$, if $u \sqsupseteq v$, then $u \sqsupseteq v$.
only sublocations verify φ
 φ is true **only within** u .

We now characterize four locational modalities in terms of $\Box$ and $\Box$:

name	formal	interpretation
pervasive	$\Box \varphi$	true all over u
weak	$\Diamond \varphi$	true in some of u
entire	$\Diamond \varphi \wedge \Box \varphi$	true within u
perfect	$\Box \varphi \wedge \Box \Box \varphi$	true all over <i>and</i> only within u



4. Further Locative Modalities

Two important observations:

1. Proposition have at most one perfect location.
2. No proposition is entirely located at a proper sublocation of its perfect location, if there is one.

- (1) is perfectly located at the union of all election days, which is not a time interval but rather a scattered union of intervals.
- (1) is *not* entirely within November 3, 2026.

Since major US presidential elections occur every four years, there are different **truthmakers** for the the proposition, which are themselves events:

- The US election on November 3, 2026.
- The US election on November 7, 2028.

Since events come with temporal locations, we may be tempted to claim:

- (1) is **weakly entirely** within Nov. 3, 2026
- (1) is **exactly located** at Nov. 3, 2026

- (1) is **weakly entirely** within Nov. 7, 2028
- (1) is **exactly located** at Nov. 7, 2028.

Exact location is not additive. Even if (1) is exactly located at both Nov. 3, 2026 *and* Nov. 7, 2028, it is **not** exactly located at the union interval.

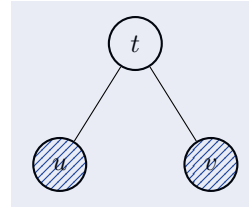
5. Exact Location is Undefinable

Main Thesis. Exact location is not definable in terms of $\Box$ and $\Box \Box$ alone.

Why? Two propositions can exemplify the same pattern of truth distribution over a locational frame, and yet differ with respect to weak entire and exact location.

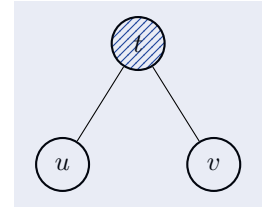
Illustration. Let p be exactly located at u, v in one case, and at t in the other:

Scenario 1



Realizers: u and v
 p is **not** exactly located at t
 p is exactly located at u
 p is exactly located at v

Scenario 2



Realizer: t
 p is exactly located at t
 p is **not** exactly located at u
 p is **not** exactly located at v

Yet, p exemplifies the same patterns of truth distributions definable in terms of $\Box$ and $\Box \Box$

To capture weak entire and exact location we must supplement our framework with a **realization relation** propositions to, which maps propositions to its truthmakers. But that takes us beyond the modal framework outlined before.

6. Conclusion

There are important **locational modalities** that take us beyond the pattern of distribution of truth across locations. Weak entire and exact location is partly a matter of what makes a proposition true, which cannot be read off a pattern of truth distribution.

Summary Table

locational modality	what it tracks	key feature
pervasive	truth all over	closed downward
weak	truth at some of	weakest
entire	truth only within	—
perfect	truth all over and only within	functional
weak entire	truth only within the perfect location of a truthmaker	
exact	truth all over and within the perfect location of a truthmaker	non-functional

References

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